UNIVERSITY MANAGEMENT SYSTEM

A Project report submitted to

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HIERANK BUSINESS SCHOOL

In partial fulfillment of the Requirements for the award of the Degree of

BACHELOR OF COMPUTER APPLICATION

by

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CERTIFICATE

This is to certify that this is the bonafide record of the Major Project entitled **"University Management System"**, submitted by **Shirashti Chaudhary – 230022810** of Bachelor of Computer Application in the partial fulfillment of the requirements for the degree of Bachelor of Computer Application during the year 2025-2026. The results embodied in this major project report have not been submitted to any other university or institute for the award of any degree or diploma.

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With regards and gratitude

SHIRASHTI CHAUDHARY – 230022810

DECLARATION

We hereby declare that the major project titled **"University management System"** submitted to Hierank Business School, affiliated Chaudhary Charan Singh University, Meerut for the award of the degree of Bachelor of Computer Application is a result of original work carried out in this project. It is further declared that the major project report or any part thereof has not been previously submitted to any University or Institute for the award of degree or diploma.

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ABSTRACT

The University Management System is a comprehensive full-stack web application developed to streamline and automate administrative, academic, and management processes within educational institutions. This project addresses the critical need for efficient digital infrastructure by implementing an integrated solution that connects various stakeholders including administrators, teachers, and students through a modern, accessible web interface.

The system is architected using **Python 3.13** for backend development and business logic implementation, **Streamlit** for rapid web interface generation, and **SQLite** for lightweight yet robust data storage and retrieval operations. The primary objective is to enhance operational efficiency, minimize manual administrative burden, and facilitate data-driven decision-making through integrated AI/ML analytics.

The application encompasses five core modules that form the backbone of university operations. The **Student Management** module maintains comprehensive student records including enrollment data, contact information, and academic progression monitoring. The **Marks Management** module tracks internal and external assessments with automatic grade calculation and performance analysis. The **Fee Management** module monitors tuition payments, pending balances, and payment status across semesters. The **Attendance Management** module calculates attendance percentages and identifies students below institutional thresholds. The **Teacher Management** module maintains faculty information, qualifications, and departmental assignments.

A key innovation of this system is the integration of **AI/ML analytics** for predictive insights including student performance forecasting, at-risk student identification, and attendance pattern analysis. Student **Portal** provides personalized dashboards where students can access their academic records, marks, fees, and attendance information in real-time, promoting transparency and self-monitoring.

The system employs **three-tier architecture** separating presentation, business logic,

and data layers, ensuring scalability, maintainability, and future extensibility. **Role-Based Access Control (RBAC)** implements distinct user roles for administrators, teachers, and students, each with appropriately restricted permissions. Security is enforced through **SHA-256 password hashing** and session management. The implementation features interactive data visualizations using **Plotly**, comprehensive Excel export capabilities, and an intuitive user interface requiring minimal training.

This full-stack solution transforms traditional manual processes into an efficient digital ecosystem, fostering an improved educational environment conducive to academic excellence and administrative effectiveness. The open-source nature of the project, with complete source code available on GitHub, promotes transparency, collaborative improvement, and educational value for the development community.

Keywords: University Management System, Full-Stack Web Application, Streamlit, Python, SQLite, Machine Learning, Role-Based Access Control, Student Portal, Data Analytics, Educational Software.

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ABBREVIATIONS

Abbreviation	Full Form
UMS	University Management System
RBAC	Role-Based Access Control
AI/ML	Artificial Intelligence / Machine Learning
CRUD	Create, Read, Update, Delete
UI/UX	User Interface / User Experience
SHA	Secure Hash Algorithm
SQL	Structured Query Language
CSV	Comma Separated Values
PDF	Portable Document Format
API	Application Programming Interface
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
DB	Database
ORM	Object-Relational Mapping
JSON	JavaScript Object Notation
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
IDE	Integrated Development Environment